Kritika Goyal

Jr. Data Scientist & GenAI Engineer
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PROFILE

GenAI engineer and data scientist who turns complex healthcare and business data into products people use every day. In under two years since completing an M.S. in Analytics (4.0 GPA), has delivered 19 production systems (30+ projects overall) for federal health agencies, national healthcare organizations, and internal teams, owning each from design through cloud deployment. Leads development of a large-scale healthcare benchmarking platform with automated scoring and AI-generated performance reports. Committed to growing others through mentoring, teaching, and community building, beginning as founding president of American University's Analytics Club.

RECOGNITION

- Google Hackathon – 2nd Place | MIT Sloan First Pitch Case Competition – 2nd Place
- Kogod Case Competition – 3rd Place | Deloitte Outstanding Award | Deloitte Applause Award

SELECTED PROJECTS

Healthcare Digital Maturity Benchmarking Platform *Python, React, AWS, Claude*

- Building an end-to-end benchmarking platform for a national healthcare industry survey, processing hundreds of thousands of survey responses across hundreds of healthcare organizations.
- Designed a what-if simulation tool that shows health system executives the smallest investment needed to raise their digital maturity level before committing budget.
- Replaced manual Excel-based scoring with an automated engine: a single raw survey export now produces every organization's results in minutes, reproducing all legacy benchmark scores exactly.
- Automated generation of long-form, per-organization performance reports (PDF and Word), including LLM-written narrative.

Enterprise RAG Document Assistant *Python, AWS Bedrock, OpenSearch, Claude*

- Built a retrieval-augmented AI assistant that answers staff questions from controlled document libraries; deployed to multiple business teams (e.g., quality procedures, contracts) from a single configurable codebase.
- Designed it to decline rather than guess: answers are grounded only in approved documents, cite their sources, and route unanswered questions to a designated contact, making every response traceable.
- Migrated retrieval from a custom-built vector search prototype to a managed AWS service, eliminating infrastructure upkeep and enabling automatic indexing of new documents.

AI-Powered Recruiting Data Pipeline *Python, AWS Bedrock, ECS Fargate, MCP*

- Built a nightly pipeline that ingests open requisitions and candidates from an applicant tracking system and uses LLMs to convert resumes into structured, searchable profiles.
- Cut LLM processing costs ~90% with no loss in quality by benchmarking models on real resumes and selecting a smaller, faster model; shared the evaluation findings with the team.
- Re-architected the pipeline for reliable multi-hour unattended runs with failure isolation, automatic retries, and monitoring alerts.
- Exposed candidate skills, tenure, and work history to recruiters and AI agents via MCP, replacing manual review of individual PDFs.

PROFESSIONAL EXPERIENCE

Revolutional (formerly Harmonia Holdings Group), McLean, VA *January 2025 – Present*

Jr. Data Scientist

- Sole owner of three production GenAI products (benchmarking platform, RAG document assistant, recruiting pipeline), from design through deployment.
- Built a central multi-agent hub that connects specialized AI agents behind a single interface.
- Took an AI-powered project management platform from initial build to production.
- Contributed to two federal health data platforms: a behavioral health data explorer integrating data from multiple federal agencies, and a clinical evidence platform linking hundreds of thousands of clinical trials to published research.
- Built or deployed additional applications for fraud detection, compliance, supply chain, and nutrition planning.
- Built the data pipelines and AWS cloud infrastructure behind all production applications.

AARP, Washington, D.C. *June 2024 – December 2024*

Analytics Intern

- Automated recurring reports, cutting manual effort 30%, and built real-time campaign dashboards for outreach teams.
- Reduced outreach costs 15% through demographic segmentation analysis.

American University – Kogod School of Business, Washington, D.C. *January 2024 – December 2024*

Graduate Teaching Assistant

- Taught labs on data cleaning and model validation for 30+ graduate students in Business Analytics and Managerial Statistics.

Deloitte, Hyderabad, India *July 2020 – August 2023*

Senior Analyst / Analyst

- Served as primary data liaison for a large-scale HR analytics transformation spanning HR, IT, and business operations.
- Automated offer management and reporting for 50,000+ candidates and employees, reducing manual effort 50%.
- Delivered dashboards that cut reporting time 40% across 20+ teams.

LEADERSHIP & COMMUNITY

- **Founding President, AU Analytics Club:** founded and led American University's analytics student community.
- **Mentor:** mentored 3+ data science interns; ran internal lunch-and-learns teaching colleagues emerging AI agent standards (MCP, A2A).
- **Educator:** Graduate Teaching Assistant in Business Analytics and Managerial Statistics, supporting 30+ graduate students.
- **Team Lead:** led a team of four analysts across 4 concurrent projects at Deloitte.

SKILLS

- **Languages:** Python, SQL, R
- **AI & GenAI:** LLMs, Claude on AWS Bedrock, Agentic Frameworks, NL-to-SQL Agents, MCP, A2A, RAG, Prompt Engineering, NLP, ML (Random Forest, SVM, Neural Networks, Gradient Boosting)
- **Cloud & Data:** AWS (ECS Fargate, RDS Postgres, S3, Athena, Lambda, CloudFront, ALB, Route 53, Bedrock, EventBridge, OpenSearch), Terraform, CloudFormation, Docker, Parquet data lakes, Databricks/Spark
- **Full Stack:** FastAPI, Flask, React, LangChain, PostgreSQL, Palantir Foundry (Certified), Power Apps, Copilot Studio
- **Visualization & Practices:** Tableau, Looker Studio, Excel; Agile/SDLC, Data Storytelling, Requirements Analysis

EDUCATION

Master of Science in Analytics — American University, Kogod School of Business *December 2024*

Washington, D.C. | GPA: 4.0 | Founding President, AU Analytics Club

B.Tech, Electronics & Communication Engineering — Malla Reddy College of Engineering & Technology *June 2020*

Hyderabad, India

CERTIFICATIONS

- Certified Palantir Foundry Aware Professional – Palantir Technologies | Databricks Fundamentals Accreditation – Databricks
- Software Engineering for Data Science – IIIT Hyderabad / Great Learning | Future of Work Institute Certificate – Deloitte